4. Nonfunctional Requirements

NFR-01: Performance
The system shall complete dependency impact analysis within 3 seconds for a process containing up to 500 activities and 1,000 dependencies under normal operating conditions.

NFR-02: Simulation Response Time
The system shall complete a simulation and comparison within 5 seconds for a process containing up to 500 activities.

NFR-03: Usability
New users shall be able to create a basic process model and run an impact analysis within 10 minutes of using the application, without programming assistance.

NFR-04: Reliability
The system shall successfully complete at least 95% of valid test scenarios during project testing.

NFR-05: Data Integrity
The system shall preserve the original process definition in 100% of tested change scenarios when users create or modify a separate scenario.

NFR-06: Error Handling
The system shall display a meaningful error message within 2 seconds when a user submits an invalid process definition.

NFR-07: Compatibility
The web application shall support the latest stable versions of Google Chrome, Microsoft Edge, and Mozilla Firefox available at the time of testing.

NFR-08: Maintainability
The application shall organize process models, graph analysis, simulation logic, database operations, and web routes into separate modules.

NFR-09: Portability
The application shall run on Windows 10 or later using Python and documented installation instructions without requiring a cloud server.

NFR-10: Data Privacy
The demonstration dataset shall contain no confidential organizational information or real personal records.

NFR-11: Data Export
The system shall export a valid process definition containing up to 500 activities in JSON format within 5 seconds under normal operating conditions.

NFR-12: Graph Visualization
The system shall display a dependency graph containing up to 200 activities in an interactive view within 5 seconds under normal operating conditions.

NFR-13: Availability
During a two-hour local demonstration, the application shall remain operational without an unexpected shutdown.

NFR-14: Database Operations
The system shall successfully save and retrieve at least 99% of valid process records during a test of 100 database operations.

NFR-15: Accessibility and Readability
The main application pages shall use readable text, clearly labeled form fields, and sufficient contrast between text and backgrounds. All primary functions shall be accessible using a keyboard.